MTH636 – Game Theory

Lecture 2: Preferences, Utility, and Normal Form Games

1 Revisiting the Beauty Contest Game

Recall the Beauty Contest Game with $N = \{1, 2, 3\}$. Each player chooses a number between 0 and 100. The target is $\frac{2}{3}$ of the average, and the player whose choice is closest to the target wins.

The set of possible **outcomes** (denoted by O) includes:

- Player 1 is the sole winner.
- Player 2 is the sole winner.
- Player 3 is the sole winner.
- Players 1 and 2 are joint winners.
- Players 1 and 3 are joint winners.
- Players 2 and 3 are joint winners.
- All are winners.

1.1 Preferences over Outcomes

Player 1 has a preference relation over these outcomes. Naturally, Player 1 prefers to win alone rather than share the win, and prefers to share the win rather than not win at all. We denote this preference relation by $\succsim_1$.

- Player 1 is the sole winner $\succ_1$ Players 1 and 2 are joint winners $\sim_1$ Players 1 and 3 are joint winners $\succ_1$ All are winners $\succ_1$ all the other outcomes.

Formally, a preference relation $\succsim_i$ is a subset of $O \times O$. When we write $(x, y) \in \succsim_i$, or simply $x \succsim_i y$, it means Player i **weakly prefers** outcome x over outcome y.

2 Utility Functions

Definition: Utility Function

Suppose O is the set of outcomes. Then a **utility function** u is a mapping from O to $\mathbb{R}$.

Definition: Representation of a Preference

A utility function u **represents** a preference relation $\succsim$ over O if for all $x, y \in O$:

$$x \succsim y \iff u(x) \geq u(y)$$

Theorem: Existence of a Utility Representation

Suppose O is finite. Then a preference relation $\succsim$ can be represented by a utility function if and only if the preference is **complete** and **transitive**.
Note: If O is infinite, a similar result holds but requires additional continuity assumptions.

3 Normal (Strategic) Form Games

Definition: Normal Form Game

A normal form (or strategic form) game is a 3-tuple $\langle N, (S_i)_{i\in N}, (u_i)_{i\in N}\rangle$ where:

- N is a finite set of players.
- S_i is the set of strategies available to player i.
- $u_i : \prod_{j\in N} S_j \to \mathbb{R}$ is the utility function for player i, which maps a strategy profile to a real-valued payoff.

3.1 Formalizing the Beauty Contest Game

Using this definition, we can formally define the Beauty Contest Game:

- $N = \{1, 2, 3\}$
- $S_1 = [0, 100]$, $S_2 = [0, 100]$, $S_3 = [0, 100]$
- Utility functions:

$$u_1(x_1, x_2, x_3) = \begin{cases} \frac{1}{|W(x_1,x_2,x_3)|} & \text{if } 1 \in W(x_1, x_2, x_3) \\ 0 & \text{otherwise} \end{cases}$$

where $W(x_1, x_2, x_3) = \arg\min_{i\in N} \left| x_i - \frac{2}{3}\bar{x} \right|$ is the set of winning players.

4 Classic 2x2 Games

Example: Prisoner's Dilemma

$N = \{P_1, P_2\}$, $S_1 = S_2 = \{C, D\}$ (Cooperate, Defect)

$P_1 \setminus P_2$	C	D
C	$(-1, -1)$	$(-5, 0)$
D	$(0, -5)$	$(-3, -3)$

In this game, players will play (D, D) because for both of them, choosing D strictly dominates choosing C.

Example: Battle of the Sexes

$N = \{\text{Man}, \text{Woman}\}$, $S_M = S_W = \{F, D\}$ (Football, Dance)

$M \setminus W$	F	D
F	(2, 1)	(0, 0)
D	(0, 0)	(1, 2)

This is a classic coordination game where players wish to coordinate on the same event but have differing preferences over which event to attend.